

FIREWALL && DNS

Première configuration

1 - on installe en premier lieu le paquet et les dépendances bind9

```
bind9-dnsutils/oldstable-security,now 1:9.18.44-1~deb12u1 amd64 [installé]  
bind9-host/oldstable-security,now 1:9.18.44-1~deb12u1 amd64 [installé]  
bind9-libs/oldstable-security,now 1:9.18.44-1~deb12u1 amd64 [installé, automatique]  
bind9-utils/oldstable-security,now 1:9.18.44-1~deb12u1 amd64 [installé, automatique]  
bind9/oldstable-security,now 1:9.18.44-1~deb12u1 amd64 [installé]
```

2 - on installe le pare feu

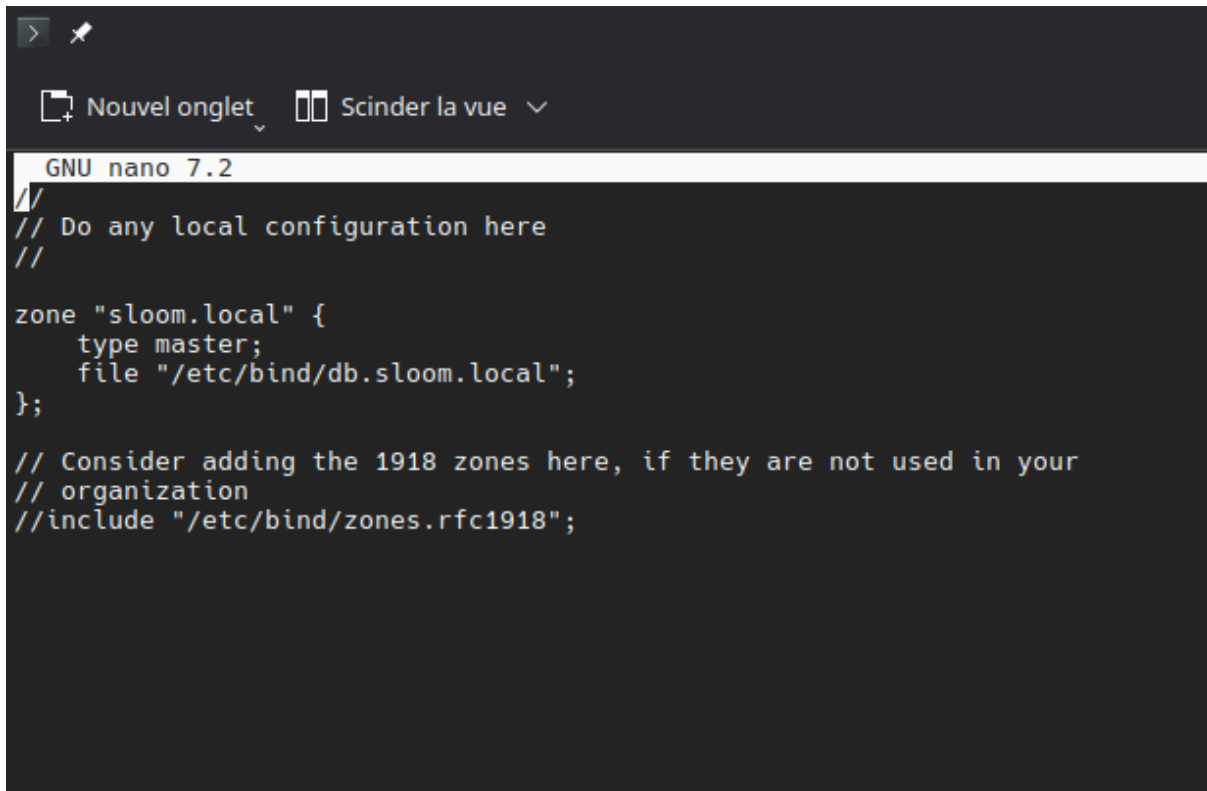
```
etudiant@debian:~$ sudo apt install ufw  
Lecture des listes de paquets... Fait  
Construction de l'arbre des dépendances... Fait  
Lecture des informations d'état... Fait
```

3 - on autorise le ssh et 53UDP pour DNS puis on active le pare feu

```
etudiant@debian:~$ sudo ufw allow 53/tcp  
Rules updated  
Rules updated (v6)  
etudiant@debian:~$ sudo ufw allow ssh  
Rules updated  
Rules updated (v6)  
etudiant@debian:~$ sudo ufw enable
```

4 - on définit le réseau sloop dans le fichier de configuration

```
etudiant@debian:~$ vim /etc/bind/named.conf.local
```



```
GNU nano 7.2
//
// Do any local configuration here
//

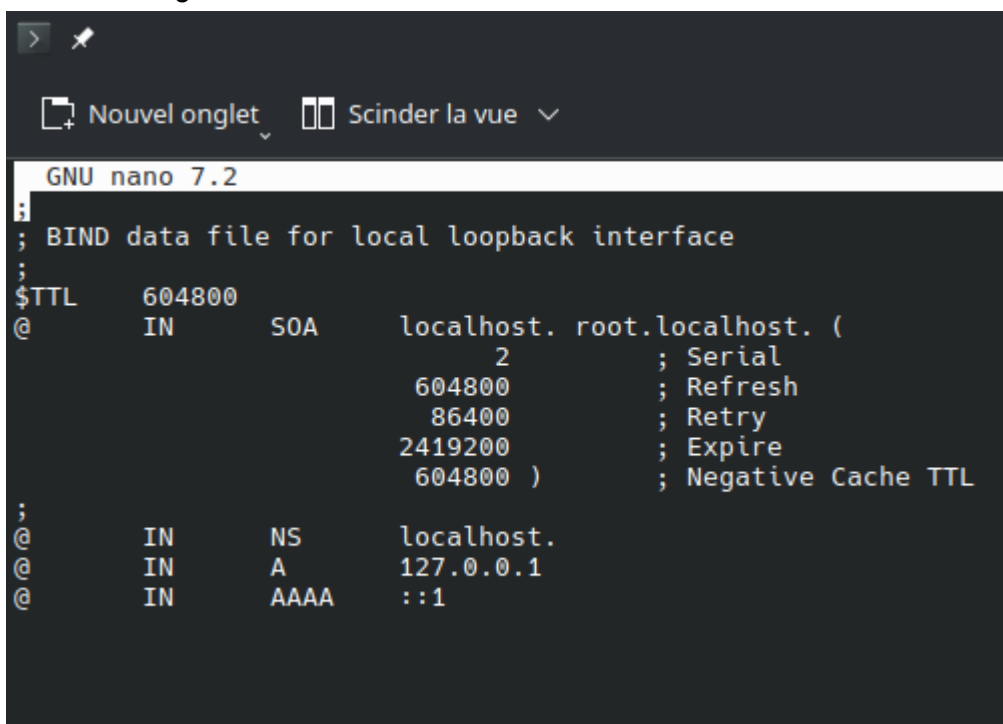
zone "sloom.local" {
    type master;
    file "/etc/bind/db.sloom.local";
};

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";
```

5 - on copie le modèle situé a `/etc/bind/db.local` et on le colle dans `/etc/bind/db.sloom.local`

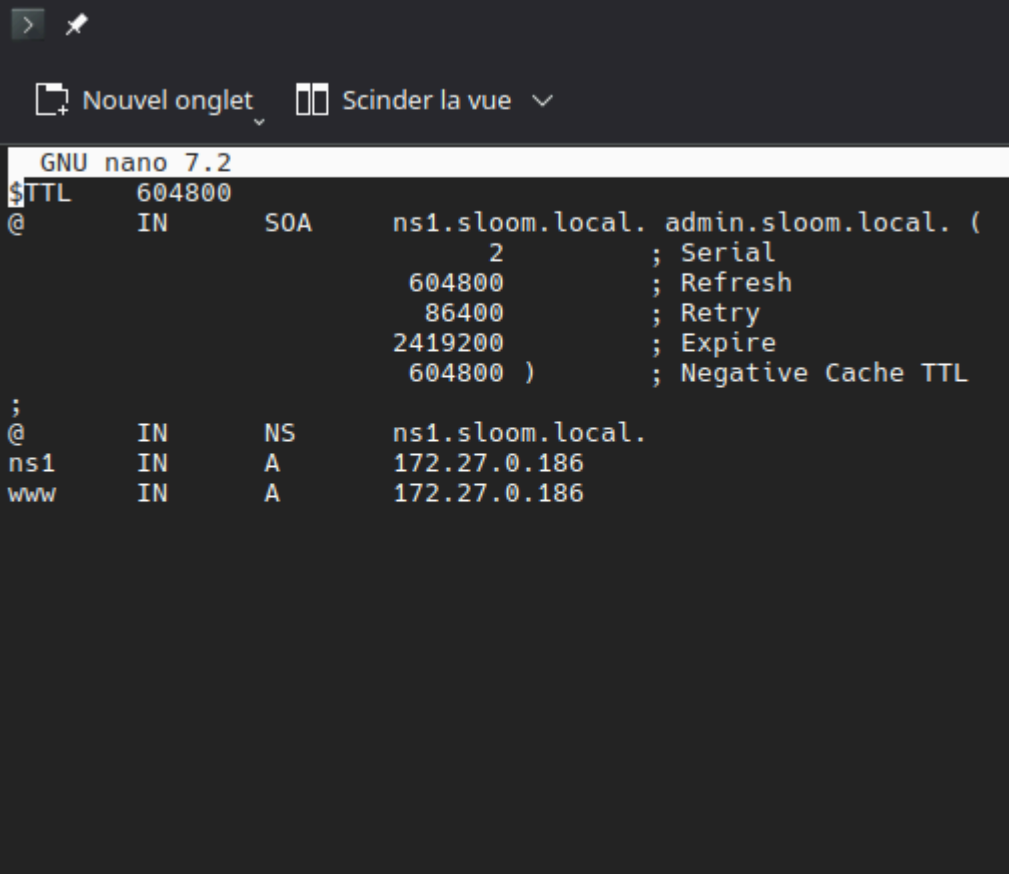
```
etudiant@debian:~$ sudo cp /etc/bind/db.local /etc/bind/db.sloom.local
```

6 - on ouvre le modèle qu'on a copié et on l'édite aux besoins de sloom
le fichier d'origine :



```
GNU nano 7.2
;
; BIND data file for local loopback interface
;
$TTL      604800
@         IN      SOA      localhost. root.localhost. (
                                2      ; Serial
                                604800  ; Refresh
                                86400   ; Retry
                                2419200 ; Expire
                                604800 ) ; Negative Cache TTL
;
@         IN      NS       localhost.
@         IN      A        127.0.0.1
@         IN      AAAA     ::1
```

après modification



```
> ✂
Nouvel onglet Scinder la vue
GNU nano 7.2
$TTL 604800
@ IN SOA ns1.sloom.local. admin.sloom.local. (
        2 ; Serial
        604800 ; Refresh
        86400 ; Retry
        2419200 ; Expire
        604800 ) ; Negative Cache TTL
;
@ IN NS ns1.sloom.local.
ns1 IN A 172.27.0.186
www IN A 172.27.0.186
```

7 - on vérifie l'existence de notre zone sloom

```
etudiant@debian:~$ sudo named-checkzone sloom.local /etc/bind/db.sloom.local
zone sloom.local/IN: loaded serial 2
OK
etudiant@debian:~$
```

8 - on relance le service

```
etudiant@debian:~$ sudo systemctl restart bind9
[sudo] Mot de passe de etudiant :
etudiant@debian:~$
```

9 - test de vérité ultime avec dig

```

etudiant@debian:~$ dig @localhost ns1.sloom.local

; <<>> DiG 9.18.44-1~deb12u1-Debian <<>> @localhost ns1.sloom.local
; (2 servers found)
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 27817
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: a8ab7277dd58c0180100000069bab419c2e2953b3db5536a (good)
;; QUESTION SECTION:
;ns1.sloom.local.                IN      A

;; ANSWER SECTION:
ns1.sloom.local.                604800  IN      A      172.27.0.186

;; Query time: 0 msec
;; SERVER: ::1#53(localhost) (UDP)
;; WHEN: Wed Mar 18 15:18:01 CET 2026
;; MSG SIZE rcvd: 88

etudiant@debian:~$

```

10 - test du domaine et de l'alias www avec nslookup

```

etudiant@debian:~$ nslookup ns1.sloom.local localhost
Server:          localhost
Address:         ::1#53

Name:   ns1.sloom.local
Address: 172.27.0.186

etudiant@debian:~$ nslookup www.sloom.local localhost
Server:          localhost
Address:         ::1#53

Name:   www.sloom.local
Address: 172.27.0.186

```

Certificats

1 - on crée les répertoires de stockages du certificat

```

etudiant@debian:~$ sudo mkdir -p /etc/apache2/ssl
[sudo] Mot de passe de etudiant :
etudiant@debian:~$

```

2 - on se déplace dans le dossier

```

etudiant@debian:~$ cd /etc/apache2/ssl
etudiant@debian:/etc/apache2/ssl$

```

3 - on génère une clé RSA de 2048 bits

```
etudiant@debian:/etc/apache2/ssl$ sudo openssl genrsa -out sloom.key 2048
etudiant@debian:/etc/apache2/ssl$
```

4 - on crée la demande de certificat

```
etudiant@debian:/etc/apache2/ssl$ sudo openssl req -new -key sloom.key -out sloom.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:FR
State or Province Name (full name) [Some-State]:SloomOrg
Locality Name (eg, city) []:Sloom City
Organization Name (eg, company) [Internet Widgits Pty Ltd]:SLOOM LTD Corp.
Organizational Unit Name (eg, section) []:DSI
Common Name (e.g. server FQDN or YOUR name) []:www.sloom.local
Email Address []:contact@sloom.fr
|
Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:Flaubert
An optional company name []:
etudiant@debian:/etc/apache2/ssl$
```

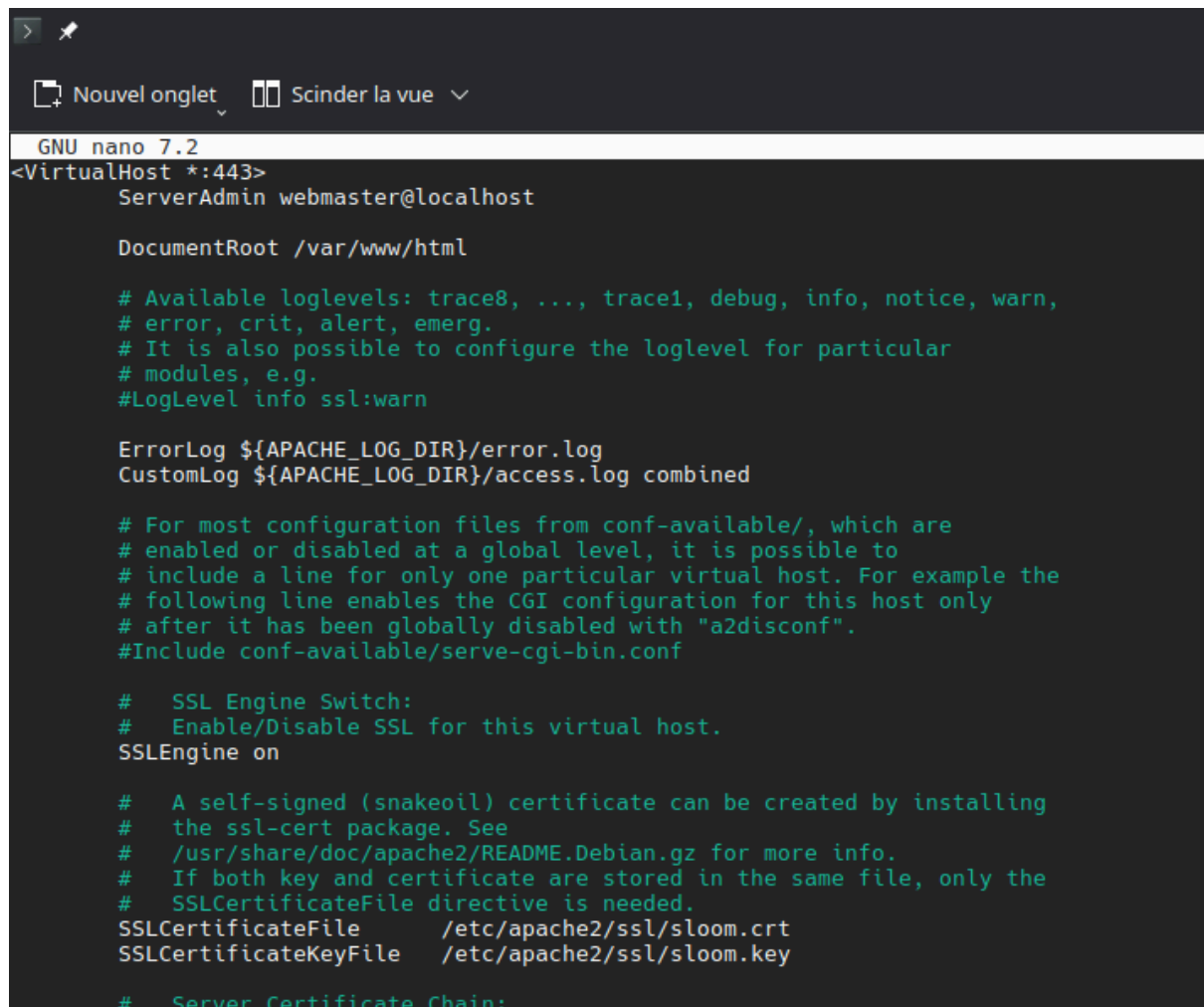
5 - on signe le certificat

```
etudiant@debian:/etc/apache2/ssl$ sudo openssl x509 -req -days 365 -in sloom.csr -signkey sloom.key -out sloom.crt
Certificate request self-signature ok
subject=C = FR, ST = SloomOrg, L = Sloom City, O = SLOOM LTD Corp., OU = DSI, CN = www.sloom.local, emailAddress = contact@sloom.fr
etudiant@debian:/etc/apache2/ssl$
```

6 - besoin d'activer le certificat

```
etudiant@debian:/etc/apache2/ssl$ sudo a2enmod ssl
[sudo] Mot de passe de etudiant :
Désolé, essayez de nouveau.
[sudo] Mot de passe de etudiant :
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
    systemctl restart apache2
etudiant@debian:/etc/apache2/ssl$
```

7 - besoin d'indiquer aux futurs connexions la localisation de ce certificat, on le fait dans le fichier suivant : `/etc/apache2/sites-available/default-ssl.conf`



```
GNU nano 7.2
<VirtualHost *:443>
    ServerAdmin webmaster@localhost

    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf

    # SSL Engine Switch:
    # Enable/Disable SSL for this virtual host.
    SSLEngine on

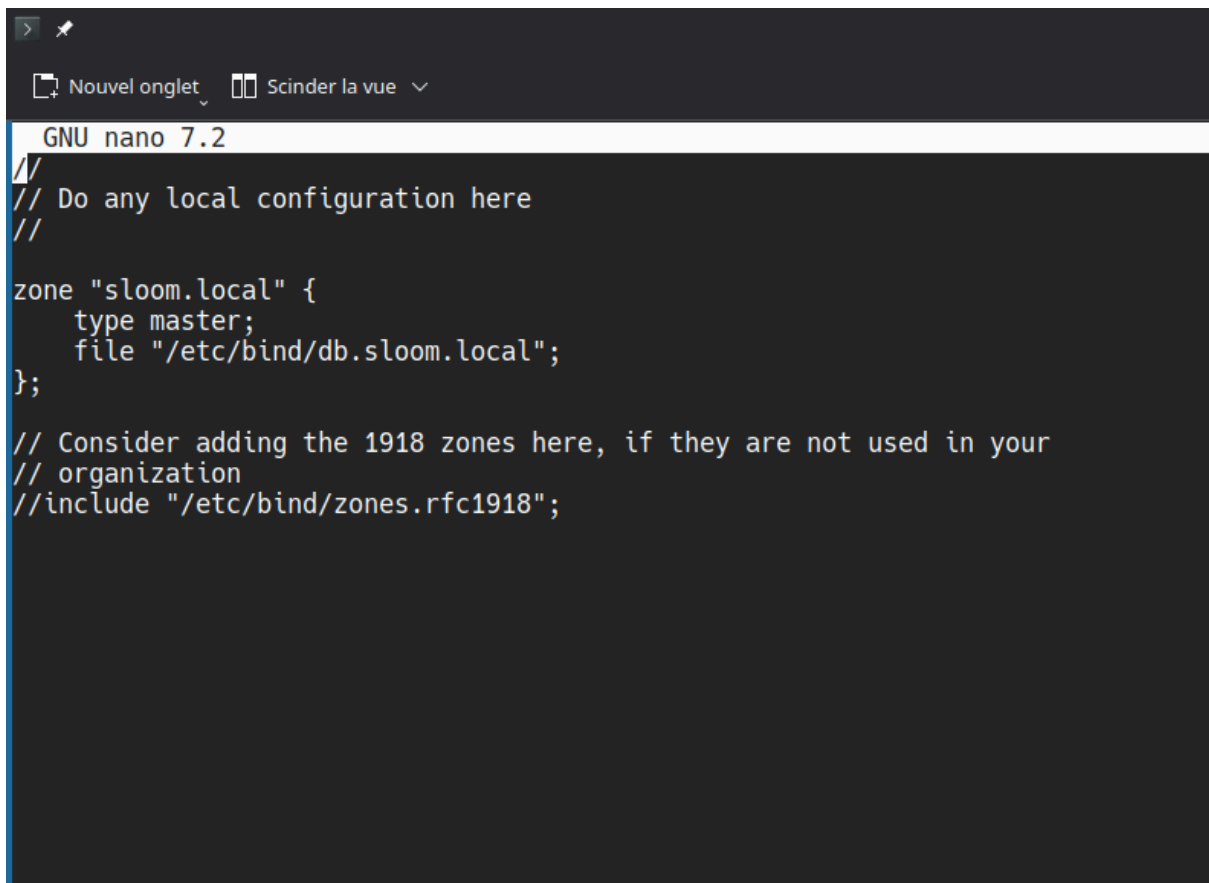
    # A self-signed (snakeoil) certificate can be created by installing
    # the ssl-cert package. See
    # /usr/share/doc/apache2/README.Debian.gz for more info.
    # If both key and certificate are stored in the same file, only the
    # SSLCertificateFile directive is needed.
    SSLCertificateFile      /etc/apache2/ssl/sloom.crt
    SSLCertificateKeyFile   /etc/apache2/ssl/sloom.key

    # Server Certificate Chain:
```

DNS

retour aux problèmes rencontrés

1 - on déclare la zone dans le fichier /etc/bind/named.conf.local

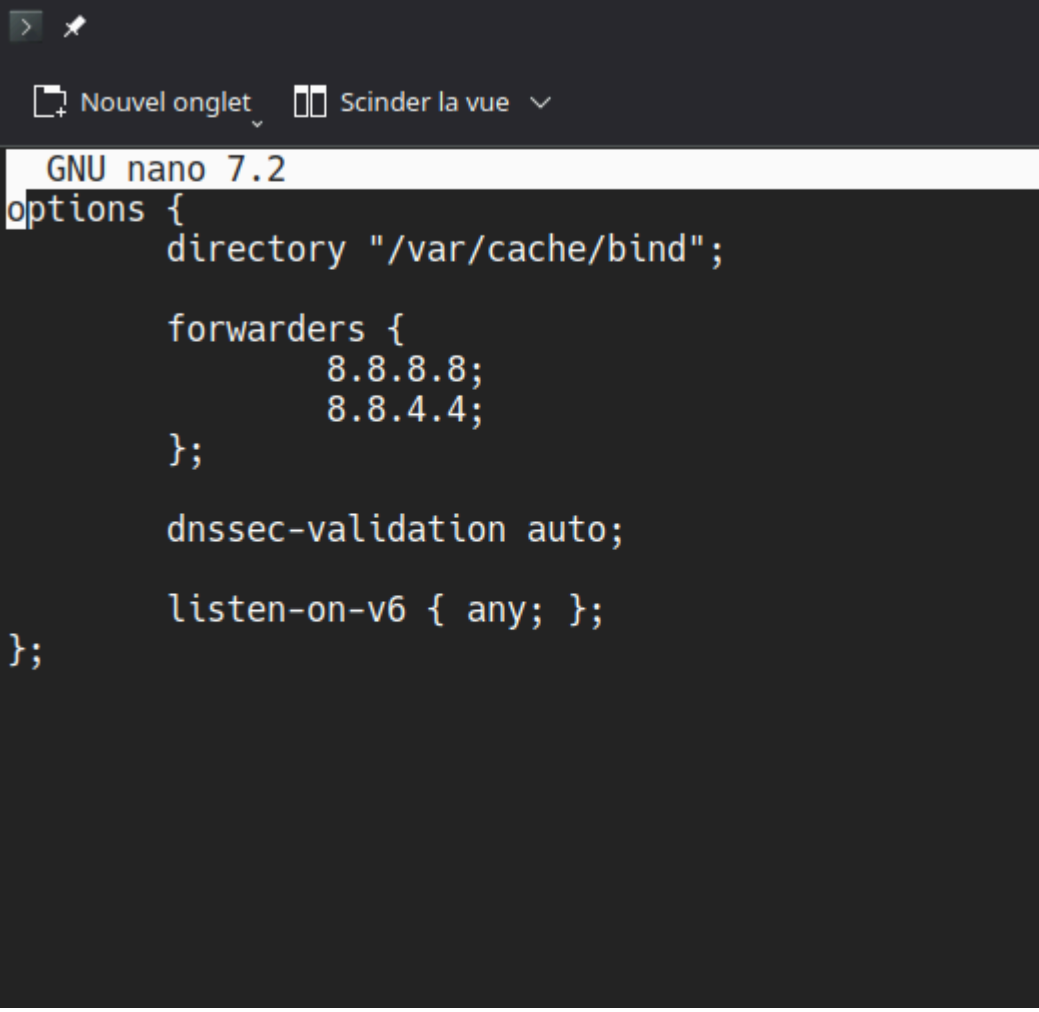


```
GNU nano 7.2
//
// Do any local configuration here
//

zone "sloom.local" {
    type master;
    file "/etc/bind/db.sloom.local";
};

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";
```

2 - nous devons corriger des erreurs de syntaxe et renseigner des serveurs tiers, ceux de google dans le fichier "/etc/bind/named.conf.options"



```
GNU nano 7.2
options {
    directory "/var/cache/bind";

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

    dnssec-validation auto;

    listen-on-v6 { any; };
};
```

3 - on intervient a nouveau sur le fichier qui contient ce qui nous permet de renvoyer une ip vers un lien situé a "/etc/bind/db.sloom.local"

```
> ✎
+ Nouvel onglet  Scinder la vue  v
GNU nano 7.2
$TTL      604800
@         IN      SOA      ns1.sloom.local. admin.sloom.local. (
                                3          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ns1.sloom.local.
@         IN      A        172.27.0.186
ns1       IN      A        172.27.0.186
www       IN      A        172.27.0.186
db        IN      A        172.27.0.186
```

4 - on vérifie que tout fonctionne correctement avec la commande “named-checkzone sloom.local /etc/bind/db.sloom.local”

```
etudiant@debian:/etc/bind$ named-checkzone sloom.local /etc/bind/db.sloom.local
zone sloom.local/IN: loaded serial 3
OK
```

elle renvoie ok donc il n’y a pas de problèmes de syntaxe

5 - moment de vérité, nous testons les différents fonctionnements du dns

- domaine racine

```
etudiant@debian:/etc/bind$ nslookup sloom.local
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   sloom.local
Address: 172.27.0.186

etudiant@debian:/etc/bind$
```

- service web

```
etudiant@debian:/etc/bind$ nslookup www.sloom.local
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   www.sloom.local
Address: 172.27.0.186

etudiant@debian:/etc/bind$
```

- la base de données

```
etudiant@debian:/etc/bind$ nslookup db.sloom.local
Server:          127.0.0.1
Address:         127.0.0.1#53

Name:   db.sloom.local
Address: 172.27.0.186

etudiant@debian:/etc/bind$
```

- test de résolution inverse
- le dns de google pour verifier la résolution des dns extérieurs

```
etudiant@debian:/etc/bind$ nslookup google.fr
Server:          127.0.0.1
Address:         127.0.0.1#53

Non-authoritative answer:
Name:   google.fr
Address: 142.250.140.94
Name:   google.fr
Address: 2a00:1450:4009:c0b::5e
```

6 - on n'oublie pas d'effectuer la zone inverse

```

etudiant@debian:/etc/bind$ cat /etc/bind/db.172.27.0
$TTL      604800
@         IN      SOA      ns1.sloom.local. admin.sloom.local. (
                                1          ; Serial
                                604800     ; Refresh
                                86400      ; Retry
                                2419200    ; Expire
                                604800 )   ; Negative Cache TTL
;
@         IN      NS       ns1.sloom.local.
186       IN      PTR      sloom.local.
186       IN      PTR      www.sloom.local.
186       IN      PTR      db.sloom.local.
etudiant@debian:/etc/bind$

```

7 - on donne son accès, on indique au fichier de comment le trouver dans /etc/bind/named.conf

```

etudiant@debian:/etc/bind$ cat /etc/bind/named.conf.local
//
// Do any local configuration here
//

zone "sloom.local" {
    type master;
    file "/etc/bind/db.sloom.local";
};

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

// zone inverse
zone "0.27.172.in-addr.arpa" {
    type master;
    file "/etc/bind/db.172.27.0";
};

```